SYSTEM READY: V1.0-STABLE

CRYPTO INTELLIGENCE BOT

Testing Phase 1: Technical Specification Baseline

📈 Multi-Timeframe RSI Engine

- **Loop Cycle:** Automated scanning every `10 minutes`.
- **Asset Coverage:** Monitors symbols in `tracked_coins`.

  BTC ETH SOL XRP ADA
- **Trigger Thresholds:** Boundary triggers at `Oversold < 25` and `Overbought > 75`.
- **Volume Filter:** Requires candle volume to exceed `1.15x` of the average of the **previous 20 candles** for that specific timeframe.
- **Deduplication:** State persistent via `rsi_state.json`.

🚀 Real-Time Intelligence Alerts

- **Breaking News:** Automated news loop checks the configured news feed every 5 minutes, deduplicates alerts, and skips stale articles after restart.
- **Exchange-Wide Liquidations:** Integrated with Binance `!forceOrder@arr`, capturing every market-wide Binance Futures liquidation.
- **Alert Logic:** Broadcasts for tracked coins when liquidation exceeds `$50,000`. Includes trend confluence (4h SMA) and custom intelligence insights.

📰 Executive Market Digest

- **Scheduling:** Configurable via UTC settings (Default: `04:00 UTC / 09:30 AM IST`).
- **Pinned Core Assets:** `BTC`, `ETH`, and `SOL` stay permanently pinned as core digest coins even if tracking list is modified.
- **Market Pulse:** Formula-driven 0-100 score weighing **Trend, RSI, Price Change, Powell Bias, and ETF Bias**.
- **Liquidation matrix:** Shows **Total Amount**, absolute **Long Value**, **Short Value**, and **Composition Ratio (%)** with status labels: `NORMAL / HIGH / EXTREME`.
- **Macro Outlook:** Includes a dedicated `Upcoming Macro (24h)` section showing high-impact events.
- **Curated News:** Caps at **Top 3** items; includes Severity Label, Source, and a "Why it matters" impact explanation.

🛠️ Full Command & Admin Suite

- **Menu Sync:** Automated Telegram menu registration via `setMyCommands`, with admin-exclusive commands hidden from public view.
- **Operational:** `/digest` (Immediate manual broadcast) and `/macro` (7-day audit checking both **this week** and **next week** sources to avoid weekend blind spots).
- **Administrative:** `/users`, `/channels`, and `/send` (Global Broadcast).
- **Access Control:** `/revoke` and `/renew` (Channel Access Management).
- **Live Config:** `/update_coins`, `/update_powell`, and `/update_etf` (Live persistent JSON modification).

🛡️ Technical Persistence & Resilience

- **Persistence Layer:** Exact file-state management for: `config.json`, `rsi_state.json`, `news_state.json`, `macro_state.json`, `last_digest.txt`.
- **Primary Storage:** SQLite database tracking `liquidations`, `alerts`, `users`, and `channels`.
- **Fail-Safe Logic:** Persistent retry count with **linear incremental backoffs** (capped at 60s).
- **Emergency Alert:** Automated **Admin Emergency DM** sent after 3 consecutive exchange connection failures.